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**Algorithm 1** Target–Evaluate–Improve (TEI)

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**Require:** Inputs: baseline agent  $A_0$ , probe set  $P$ , dimensions  $D$

**Require:** Hyperparams: budget  $N$ , batch size  $b$

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1:  $L \leftarrow \{(A_0, S(A_0))\}$ 
2: for  $t = 1, \dots, \lceil N/b \rceil$  do
3:    $A^* \leftarrow \arg \max_{(A, S) \in L} S$ 
4:    $d_t^* \leftarrow \arg \min_{d \in D} s_d(A^*)$   $\triangleright$  TARGET
5:    $C_t \leftarrow \text{PROPOSE}(A^*, d_t^*, b; L)$ 
6:   for  $c \in C_t$  do
7:     if  $\text{ASTGATE}(c)$  then
8:       score  $c$  on  $P$  by Eq. 1  $\triangleright$  EVALUATE
9:       add  $(A_c, S(c))$  to  $L$ 
10:    end if
11:  end for
12: end for
13:  $c^* \leftarrow \arg \max_{(A, S) \in L} S$ 
14: return  $A_{c^*}$  if  $\text{GATE}(c^*, \text{ref})$  else  $A_0$   $\triangleright$  IMPROVE
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**Algorithm 2** Do-no-harm deployment gate

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**Require:** paired probe-query scores  $S_i(\cdot)$  on  $P$ ,  
 $i = 1, \dots, n$ ; deployment reference  $\text{ref}$

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1: function  $\text{GATE}(c^*, \text{ref})$ 
2:    $\epsilon \leftarrow 10^{-9}$ 
3:    $W \leftarrow \#\{i : S_i(c^*) > S_i(\text{ref}) + \epsilon\}$ 
4:    $L \leftarrow \#\{i : S_i(c^*) < S_i(\text{ref}) - \epsilon\}$ 
5:   if  $\bar{S}_{c^*} < \bar{S}_{\text{ref}}$  then return FALSE
6:   if  $n < 3$  then return  $(L = 0)$   $\triangleright$  small  $n$ 
7:   return  $(L \leq W)$ 
8: end function
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Eq. 1  $S(c) = \frac{1}{4} \sum_{d \in D} \text{clamp}(s_d(c)), \quad \text{clamp}(s) = \min(\max(s, 0), 0.999).$